

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: Lab 1

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Student ID: 41972028

GitHub Repository Link: https://github.com/LilyGreat101/Lab_1A.git

Date Submitted: 20 June 2026

1. AI Use Summary

Did you use any AI tool for this lab? :Yes

Estimated percentage of the work influenced by AI :20%

Did you attach evidence of AI use where applicable? :Yes

2. Details of AI Use

| Name of AI Tool Used | Purpose for Using the Tool | Prompt or Instruction Given to the Tool | Part(s) of the Work Influenced | How I Verified, Edited, or Improved the Output |

| ChatGPT | Debugging and fixing search algorithm implementations | “fix WeightedGridProblem constructor error”, “why is my PriorityQueue crashing” | I ran all code in my notebook, fixed errors step-by-step, tested assertions, and adjusted code based on lab requirements |

ChatGPT | Understanding AI search concepts and heuristics | “simple explanation of Manhattan vs Euclidean heuristic”, “why is A* better than Greedy” | Reflection questions (Section 16) | I made sure I understood the concepts and applied my thinking |

ChatGPT | README and GitHub setup guidance | “how to write README for AI search lab”, “how to push code to GitHub repo” | README.md and repository setup | I edited README to match my project and verified Git commands on my system |

3. Attachment of AI Output Evidence

| AI-generated draft | No

| Prompt history | No

| Screenshot of AI interaction | Yes | screenshots.png |

| Exported AI conversation | No

| AI-generated code snippet | No

| Revised version showing student input | No

4. Student Declaration

I declare that:

| The submitted work is my own work. | Yes |

| Any use of AI tools has been clearly declared in this form. | Yes |

| The prompts, instructions, and AI-generated outputs have been disclosed where applicable. | Yes |

| I reviewed, tested, edited, and improved any AI-generated content before submission. | Yes |

| My AI usage does not exceed 25% of the entire work. | Yes |

| I understand that undeclared or excessive AI use may be treated as academic misconduct. | Yes |

Student Signature:Lily Great Ayensu

Date:20 June 2026

ChatGPT

less PriorityQueue:
"""A min-priority queue of (priority, node) pairs built on heapq."""

def __init__(self):
 self._heap: List[Tuple[float, int, Node]] = []
 self._counter = itertools.count()

def push(self, priority: float, node: Node) -> None:
 heapq.heappush(self._heap, (priority, next(self._counter), node))

def pop(self) -> Node:
 Show more

Upgrade

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You've got a typo + missing imports problem risk here, but the main bug is simple:

1. Typo in your code

You wrote:

```
Python  
class PriorityQueue:
```

That should be:

```
Python  
class PriorityQueue:
```

2. Missing imports (likely needed)

Your code uses `heapq`, `itertools`, `List`, `Tuple`, and `Node`.
So you probably need:

```
Python  
import heapq  
import itertools  
from typing import List, Tuple
```


(And `Node` should already exist in your project)

Ask anything

ChatGPT can make mistakes. Check important info.

ChatGPT

If you want

Paste the exact error text here and I'll pinpoint it in one shot instead of guessing.

TypeError

Traceback (most recent call last)
/tmp/pykernel_33462/34E4656363.py in <call line 0>() :
11)
12
--> 13 wtest_problem = WeightedGridProblem(
14 wtest_grid, start=(0, 0), goal=(2, 2), terrain_costs=wtest_costs
15)

TypeError: WeightedGridProblem.__init__() got multiple values for
argument 'terrain_costs'

This error is actually very specific:
`got multiple values for argument 'terrain_costs'`

What it means (simple)

You are passing `terrain_costs` twice — once positionally and once as a keyword.
So Python is confused like:
"Did you give me terrain_costs here... or here?"

Your line causing the issue

```
Python  
wtest_problem = WeightedGridProblem(  
    wtest_grid, start=(0, 0), goal=(2, 2), terrain_costs=wtest_costs  
)
```

The problem is:

🔥 In your class, `wtest_grid` is probably already being interpreted as `terrain_costs` or another positional argument.

Ask anything

ChatGPT can make mistakes. Check important info.

when it says to show output am i supposed to run it all

Yes. When a lab says "show output", your lecturer usually wants proof that the code actually runs and produces results.

For this notebook, after you've completed the TODO sections, you should run the relevant cells and leave the outputs visible:

- ☒ Run the GridProblem self-check cell and show:

```
Python
GridProblem self-check passed.
```

- ☒ Run the section:

```
Python
results = [
    bfs.search(problem),
    dfs.search(problem),
    dis.search(problem, limit=10),
    ids.search(problem, max_depth=30),
]

show_results(results)
```

and leave the DataFrame/table output visible.

- ☒ Run the visualization cells:

```
Python
plot_path(...)
```

and leave the generated path images visible.

- ☒ Run the custom map sections and leave their result tables visible.

ChatGPT

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🔗 Share ...

how do i open an ipynb file i downloaded on colab

To open a downloaded `.ipynb` notebook in Google Colab:

Method 1: Upload directly to Colab

1. Go to Google Colab »
2. Click File → Upload notebook.
3. Select your downloaded `.ipynb` file.
4. Colab will open it automatically.

Method 2: Drag and drop

1. Open Google Colab »
2. Drag the `.ipynb` file from your computer into the browser window.
3. Colab uploads and opens it.

Method 3: From Google Drive

1. Upload the `.ipynb` file to your Google Drive.
2. Right-click the file.
3. Choose Open with → Google Colaboratory.
 - If you don't see Colaboratory, click Connect more apps and search for "Colaboratory".

If the notebook won't open

- Make sure the file extension is `.ipynb`.
- If it was downloaded as a ZIP file, extract it first.
- If it's corrupted, try opening it with a text editor and check that it starts with something like:

```
JSON
{
  "cells": [
```

If you upload the notebook here, I can also inspect it and help you run it in Colab.

🔗 📄 📌 🗑️ ⋮

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